

Vector & 3D

DPP-1

- Q.1 A (1, -1, -3), B (2, 1, -2) & C (-5, 2, -6) are the position vectors of the vertices of a triangle ABC. The length of the bisector of its internal angle at A is :
(A) $\sqrt{10}/4$ (B) $3\sqrt{10}/4$ (C) $\sqrt{10}$ (D) none
- Q.2 Let $\vec{r} = \vec{a} + \lambda \vec{l}$ and $\vec{r} = \vec{b} + \mu \vec{m}$ be two lines in space where $\vec{a} = 5\hat{i} + \hat{j} + 2\hat{k}$, $\vec{b} = -\hat{i} + 7\hat{j} + 8\hat{k}$, $\vec{l} = -4\hat{i} + \hat{j} - \hat{k}$ and $\vec{m} = 2\hat{i} - 5\hat{j} - 7\hat{k}$ then the p.v. of a point which lies on both of these lines, is
(A) $\hat{i} + 2\hat{j} + \hat{k}$ (B) $2\hat{i} + \hat{j} + \hat{k}$ (C) $\hat{i} + \hat{j} + 2\hat{k}$ (D) non existent as the lines are skew
- Q.3 P, Q have position vectors $\vec{a}$ & $\vec{b}$ relative to the origin 'O' & X, Y divide $\vec{PQ}$ internally and externally respectively in the ratio 2 : 1. Vector $\vec{XY} =$
(A) $\frac{3}{2}(\vec{b} - \vec{a})$ (B) $\frac{4}{3}(\vec{a} - \vec{b})$ (C) $\frac{5}{6}(\vec{b} - \vec{a})$ (D) $\frac{4}{3}(\vec{b} - \vec{a})$
- Q.4 Let $\vec{p}$ is the p.v. of the orthocentre & $\vec{g}$ is the p.v. of the centroid of the triangle ABC where circumcentre is the origin. If $\vec{p} = K \vec{g}$, then K =
(A) 3 (B) 2 (C) 1/3 (D) 2/3
- Q.5 A vector $\vec{a}$ has components 2p & 1 with respect to a rectangular cartesian system. The system is rotated through a certain angle about the origin in the counterclockwise sense. If with respect to the new system, $\vec{a}$ has components p + 1 & 1 then ,
(A) p = 0 (B) p = 1 or p = - 1/3
(C) p = - 1 or p = 1/3 (D) p = 1 or p = - 1
- Q.6 The number of vectors of unit length perpendicular to vectors $\vec{a} = (1, 1, 0)$ & $\vec{b} = (0, 1, 1)$ is:
(A) 1 (B) 2 (C) 3 (D) ∞
- Q.7 Four points A(+1, -1, 1) ; B(1, 3, 1) ; C(4, 3, 1) and D(4, -1, 1) taken in order are the vertices of
(A) a parallelogram which is neither a rectangle nor a rhombus
(B) rhombus
(C) an isosceles trapezium
(D) a cyclic quadrilateral.
- Q.8 Let α, β & γ be distinct real numbers. The points whose position vector's are $\alpha \hat{i} + \beta \hat{j} + \gamma \hat{k}$; $\beta \hat{i} + \gamma \hat{j} + \alpha \hat{k}$ and $\gamma \hat{i} + \alpha \hat{j} + \beta \hat{k}$
(A) are collinear (B) form an equilateral triangle
(C) form a scalene triangle (D) form a right angled triangle
- Q.9 If the vectors $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = -\hat{i} + 3\hat{j} + 4\hat{k}$ & $\vec{c} = 4\hat{i} - 2\hat{j} - 6\hat{k}$ constitute the sides of a ΔABC , then the length of the median bisecting the vector $\vec{c}$ is
(A) $\sqrt{2}$ (B) $\sqrt{14}$ (C) $\sqrt{74}$ (D) $\sqrt{6}$

- Q.10 P be a point interior to the acute triangle ABC. If $\overrightarrow{PA} + \overrightarrow{PB} + \overrightarrow{PC}$ is a null vector then w.r.t. the triangle ABC, the point P is, its
 (A) centroid (B) orthocentre (C) incentre (D) circumcentre
- Q.11 A vector of magnitude 10 along the normal to the curve $3x^2 + 8xy + 2y^2 - 3 = 0$ at its point P(1, 0) can be
 (A) $6\hat{i} + 8\hat{j}$ (B) $-8\hat{i} + 3\hat{j}$ (C) $6\hat{i} - 8\hat{j}$ (D) $8\hat{i} + 6\hat{j}$
- Q.12 Consider the points A, B and C with position vectors $(-2\hat{i} + 3\hat{j} + 5\hat{k})$, $(\hat{i} + 2\hat{j} + 3\hat{k})$ and $7\hat{i} - \hat{k}$ respectively.
 Statement-1: The vector sum, $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \vec{0}$
because
 Statement-2: A, B and C form the vertices of a triangle.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false. (D) Statement-1 is false, statement-2 is true.